

**Project Design Phase-II
Technology Stack (Architecture & Stack)**

Date	23 June 2026
Team ID	
Project Name	An employee requests installation of licensed software through the Service Catalog
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Service Portal	ServiceNow Service Portal, HTML, CSS, JavaScript
2.	Application Logic-1	Logic for request submission and processing in the application	ServiceNow Workflow Engine
3.	Application Logic-2	Logic for validation and automated backend actions	ServiceNow Business Rules
4.	Application Logic-3	Logic for approvals and task automation	ServiceNow Flow Designer / Approval Engine
5.	Database	Data storage for requests, items, tasks and user information	ServiceNow Database (REQ, RITM, SCTASK Tables)
6.	Cloud Database	Database service hosted on cloud	ServiceNow Cloud Platform
7.	File Storage	Storage for configuration files, update sets and attachments	ServiceNow Update Sets / XML Storage
8.	External API-1	Purpose of external API used for notifications and communication	ServiceNow Email Notification Service
9.	External API-2	Purpose of external API for future integrations (e.g., license system)	REST API Integration
10.	Machine Learning Model	Purpose of machine learning model for future enhancements (e.g., software recommendations)	AI/ML Recommendation Model
11.	Infrastructure (Server / Cloud)	Application deployment on cloud infrastructure	ServiceNow Cloud Infrastructure

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used in the application.	• ServiceNow Platform (built on open architecture) • jQuery, AngularJS (used in Service Portal) • Bootstrap • Glide APIs
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	• Role-Based Access Control (RBAC) • Access Control Rules (ACL) • Multi-factor Authentication (MFA) • Data Encryption (in transit & at rest) • OWASP Top 10 Compliance • Audit Logs & System Security Management
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services).	• ServiceNow is a multi-tenant cloud platform. • Horizontally scalable to support more users and requests. • Modular architecture using Business Rules, Workflows, and Script Includes.
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	• ServiceNow runs on highly available cloud infrastructure. • 99.9% platform uptime SLA. • Data centers distributed across regions. • Automatic failover and redundancy.
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	• Optimized Business Rules and Workflows. • Use of Caching Mechanism. • Indexed Tables for faster retrieval. • Efficient Query Performance. • CDN for static content delivery.